

Derivative of exponential function

Find the derivative of the **exponential** function

$$f(x) = a^x, a \in \mathbb{R}, 0 < a \neq 1$$

Using the **definition** of derivatives via limits

$$\frac{df}{dx}(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

Calculating the variation of the function

Using the definition of **variation** of a function

$$\frac{\Delta y}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Being the function $f(x) = a^x$

$$\frac{\Delta y}{\Delta x} = \frac{a^{x+\Delta x} - a^x}{\Delta x}$$

Factorizing a^x , we then have the following

$$\frac{\Delta y}{\Delta x} = a^x \frac{a^{\Delta x} - 1}{\Delta x}$$

Applying the limit to the variation

Applying the limit to $\frac{\Delta y}{\Delta x}$

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = a^x \lim_{\Delta x \rightarrow 0} \frac{a^{\Delta x} - 1}{\Delta x}$$

The limit displayed above is a fundamental one

$$\lim_{\Delta x \rightarrow 0} \frac{a^{\Delta x} - 1}{\Delta x} = \ln(a)$$

Calculating the limit

Therefore

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = a^x \lim_{\Delta x \rightarrow 0} \frac{a^{\Delta x} - 1}{\Delta x} = a^x \ln(a)$$

And the result of the derivative of the function $f(x) = a^x$ is

$$\frac{df}{dx}(x) = a^x \ln(a)$$



The fundamental limit

The fundamental limit...

$$\lim_{\Delta x \rightarrow 0} \frac{a^{\Delta x} - 1}{\Delta x} = \ell n(a)$$

The Euler number and a fundamental limit

The limit

$$\lim_{\Delta x \rightarrow 0} \frac{a^{\Delta x} - 1}{\Delta x} = \ln(a)$$

Can be derived by using the definition of the number e

$$e = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x \approx 2.7182818284$$

The theorem and the proof

Let us state our theorem

$$\lim_{\Delta x \rightarrow 0} \frac{a^{\Delta x} - 1}{\Delta x} = \ell n(a)$$

Proof:

Supposing $a > 0$ and $a \neq 1$ and defining

$$b = a^{\Delta x} - 1$$

Which leads to the following

$$\Delta x = \frac{\ell n(1 + b)}{\ell n(a)}$$

The theorem and the proof (cont.)

Notice that

$$\frac{a^{\Delta x} - 1}{\Delta x} = b \frac{\ln(a)}{\ln(1 + b)}$$

With further algebraic manipulation

$$b \frac{\ln(a)}{\ln(1 + b)} = \frac{\ln(a)}{\ln(1 + b)^{\frac{1}{b}}}$$

therefore

$$\lim_{\Delta x \rightarrow 0} \frac{a^{\Delta x} - 1}{\Delta x} = \lim_{b \rightarrow 0} \frac{\ln(a)}{\ln(1 + b)^{\frac{1}{b}}}$$

The theorem and the proof (QED)

Notice that

$$\lim_{b \rightarrow 0} \frac{\ln(a)}{\ln(1+b)^{\frac{1}{b}}} = \frac{\ln(a)}{\ln \left[\lim_{b \rightarrow 0} (1+b)^{\frac{1}{b}} \right]}$$

And the denominator of the above equation equals one

$$\ln \left[\lim_{b \rightarrow 0} (1+b)^{\frac{1}{b}} \right] = \ln(e) = 1$$

Therefore, as we intended to show (QED)

$$\lim_{\Delta x \rightarrow 0} \frac{a^{\Delta x} - 1}{\Delta x} = \lim_{b \rightarrow 0} \frac{\ln(a)}{\ln(1+b)^{\frac{1}{b}}} = \ln(a)$$



Caveats and challenge

Another theorem and equivalence of limits

$$e = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = \lim_{w \rightarrow 0} (1 + w)^{\frac{1}{w}}$$

A theorem of limits

$$\lim_{b \rightarrow 0} \frac{\ln(a)}{\ln(1 + b)^{\frac{1}{b}}} = \frac{\ln(a)}{\ln \left[\lim_{b \rightarrow 0} (1 + b)^{\frac{1}{b}} \right]}$$